

Network project

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Project Overview

We are pleased to present this comprehensive documentation for our network design project, developed using **Cisco Packet Tracer**. The project aims to simulate and design a modern, scalable enterprise network that adheres to the standard **Hierarchical Engineering** model to ensure **optimal performance, high reliability, and enhanced security**. The design focuses on implementing advanced **routing concepts, network security**, and high availability (Redundancy) through protocols such as **OSPF, HSRP, and EtherChannel** and secures access using **SSH and AAA**.

The network design includes multiple layers following industry standards:

The designed network consists of three main layers: the **Core, Distribution, and Access layers**. Routers and Multilayer Switches were utilized to implement crucial tasks such as **Inter-VLAN Routing**, in addition to efficiently segmenting the network into multiple **Virtual Local Area Networks (VLANs)**.

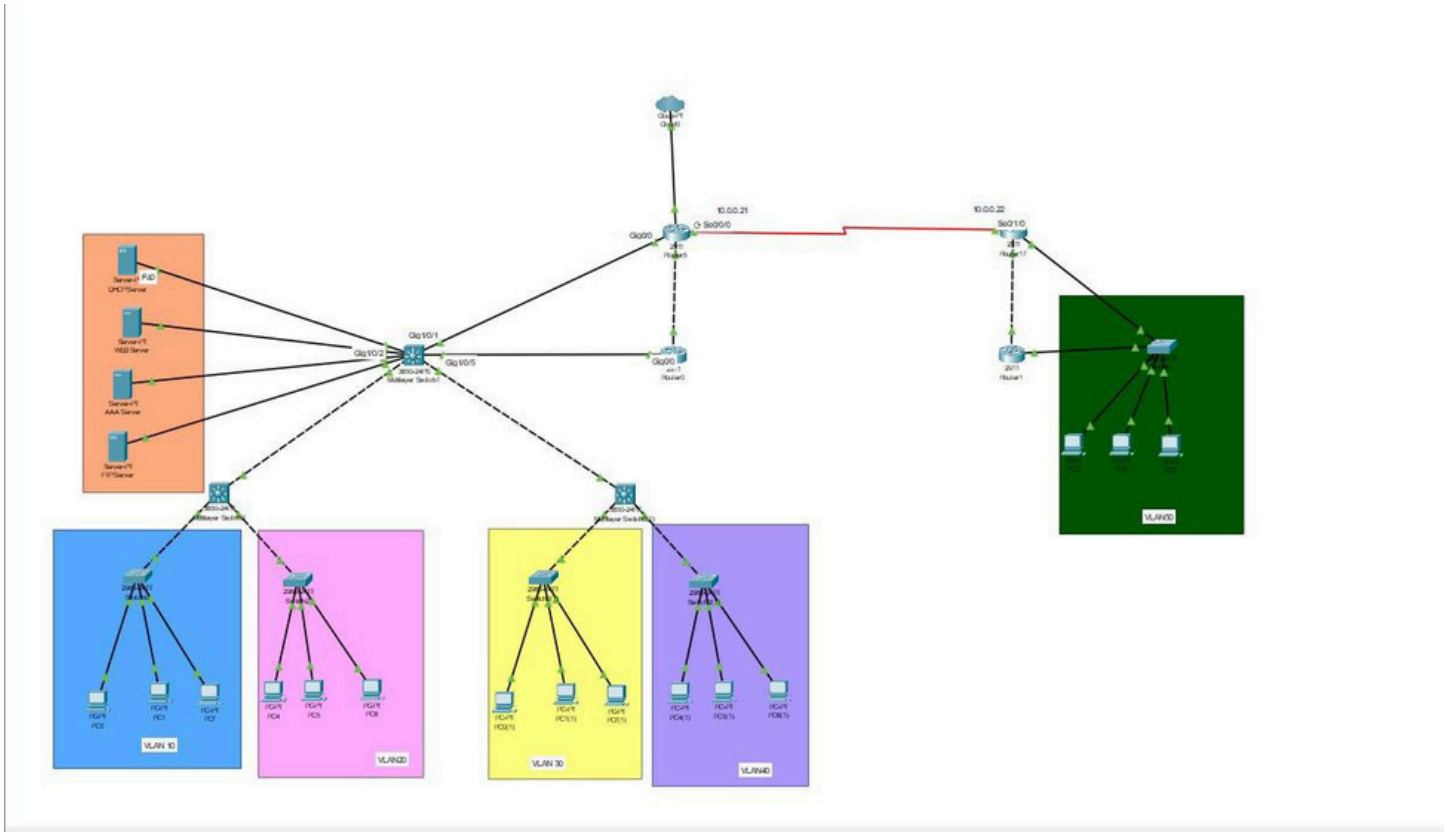
This documentation aims to explain the details of the infrastructure, the IP addressing scheme, and the key configuration processes that ensure the network operates as an integrated and effective unit.

Key technologies and configurations implemented in this project include:

The network was designed to meet the following requirements:

1. Use variable subnetting.
2. Use OSPF as the routing protocol.
3. Implement VLANs and inter-VLAN routing using ROAS.
4. Use a standalone DHCP server for network configuration.
5. Configure AAA for authentication with a local admin user.
6. Set the privilege exec mode password to the group number.
7. Use a web server with a public IP (Static NAT).
8. Configure EtherChannel for switch redundancy.
9. Use HSRP for router redundancy.
10. Backup all configurations to an FTP server.
11. Enable SSH on at least one router.

Topology Design

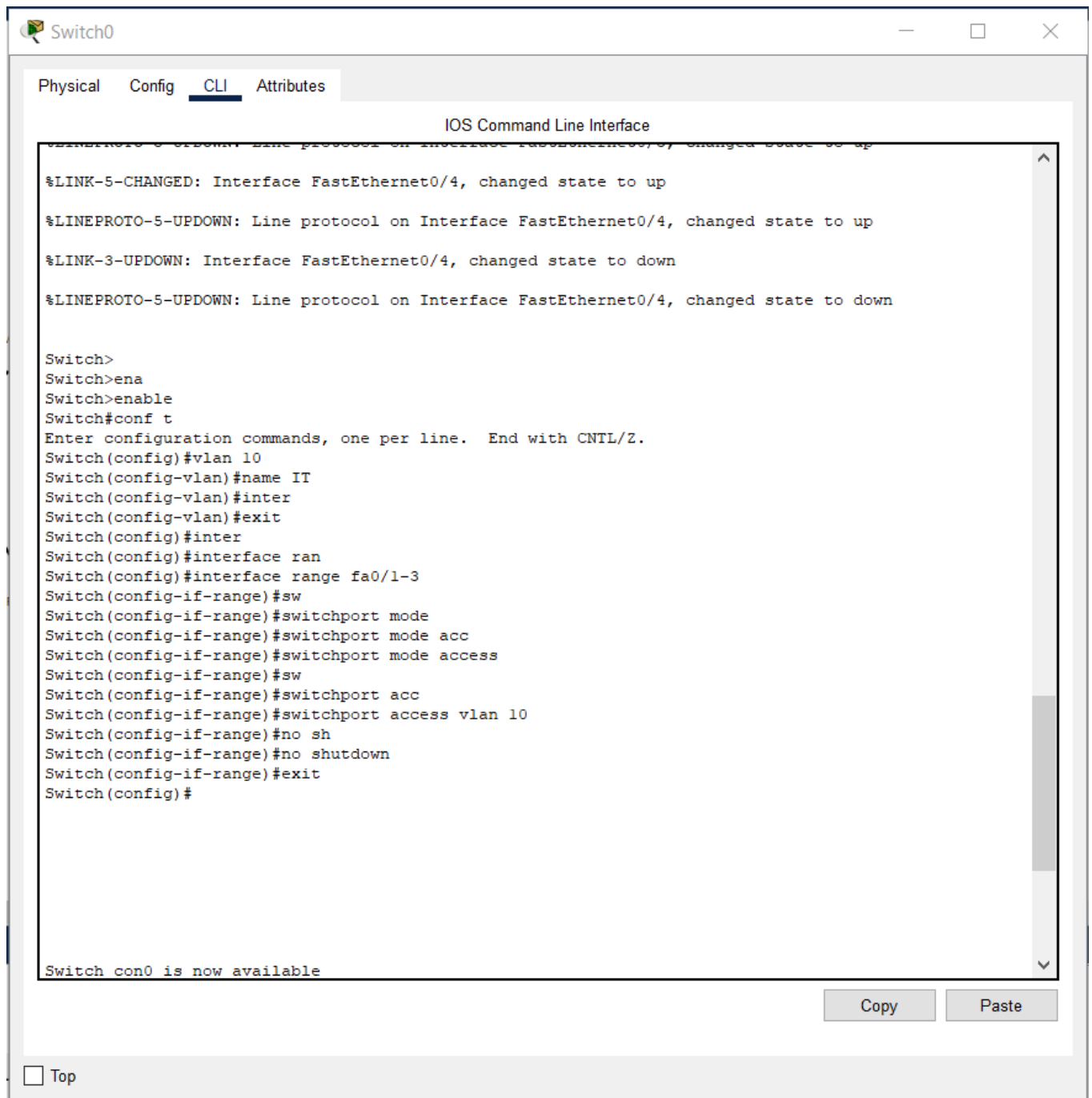


Variable Subnetting

Vlan_no	Network	Gateway/First	Last	Subnet mask	broadcast
10	192.168.5.0/26	192.168.5.1	192.168.5.62	255.255.255.192 (/26)	192.168.5.63
20	192.168.5.64/26	192.168.5.65	192.168.5.126	255.255.255.192 (/26)	192.168.5.127
30	192.168.5.128/28	192.168.5.129	192.168.5.142	255.255.255.240 (/28)	192.168.5.143
40	192.168.5.144/28	192.168.5.145	192.168.5.158	255.255.255.240 (/28)	192.168.5.159
50	192.168.5.160/28	192.168.5.161	192.168.5.174	255.255.255.240 (/28)	192.168.5.175

Implementation

- Switchport mode



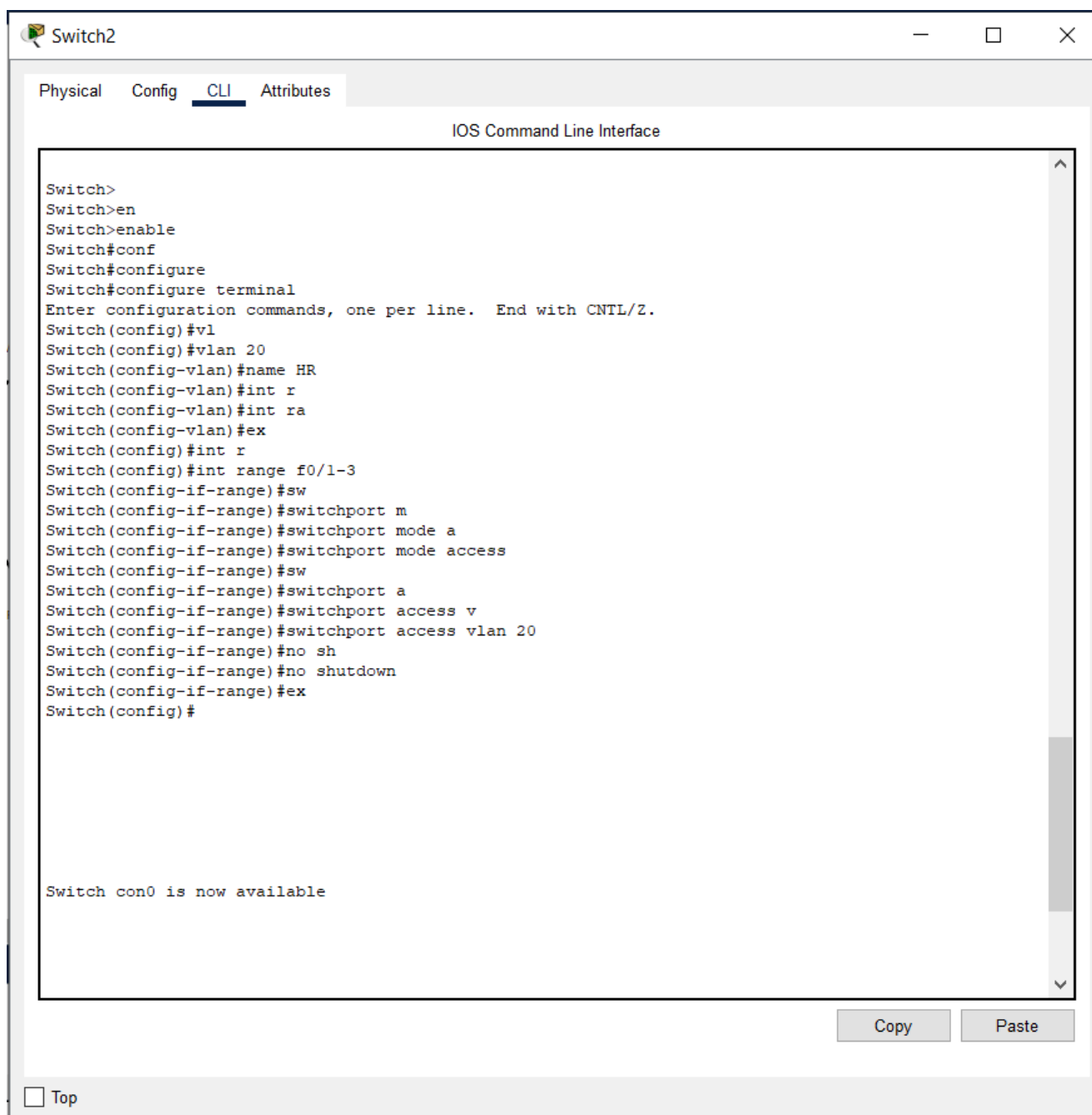
The screenshot shows a web-based interface for configuring a network switch, titled "Switch0". The interface has tabs for "Physical", "Config", "CLI", and "Attributes", with "CLI" currently selected. The main area displays the "IOS Command Line Interface" with a scrollable text box containing the following text:

```
%LINK-5-CHANGED: Interface FastEthernet0/4, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to up
%LINK-3-UPDOWN: Interface FastEthernet0/4, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to down

Switch>
Switch>ena
Switch>enable
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name IT
Switch(config-vlan)#inter
Switch(config-vlan)#exit
Switch(config)#inter
Switch(config)#interface ran
Switch(config)#interface range fa0/1-3
Switch(config-if-range)#sw
Switch(config-if-range)#switchport mode
Switch(config-if-range)#switchport mode acc
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#sw
Switch(config-if-range)#switchport acc
Switch(config-if-range)#switchport access vlan 10
Switch(config-if-range)#no sh
Switch(config-if-range)#no shutdown
Switch(config-if-range)#exit
Switch(config)#

Switch con0 is now available
```

At the bottom right of the CLI window, there are "Copy" and "Paste" buttons. Below the CLI window, there is a "Top" link with a small square icon next to it.



• EthreChannel

link aggregation technology that provides increased bandwidth capacity (Bandwidth Aggregation) and, crucially, redundancy. If one physical link in the bundle fails, the remaining links continue to pass data without network interruption

IOS Command Line Interface

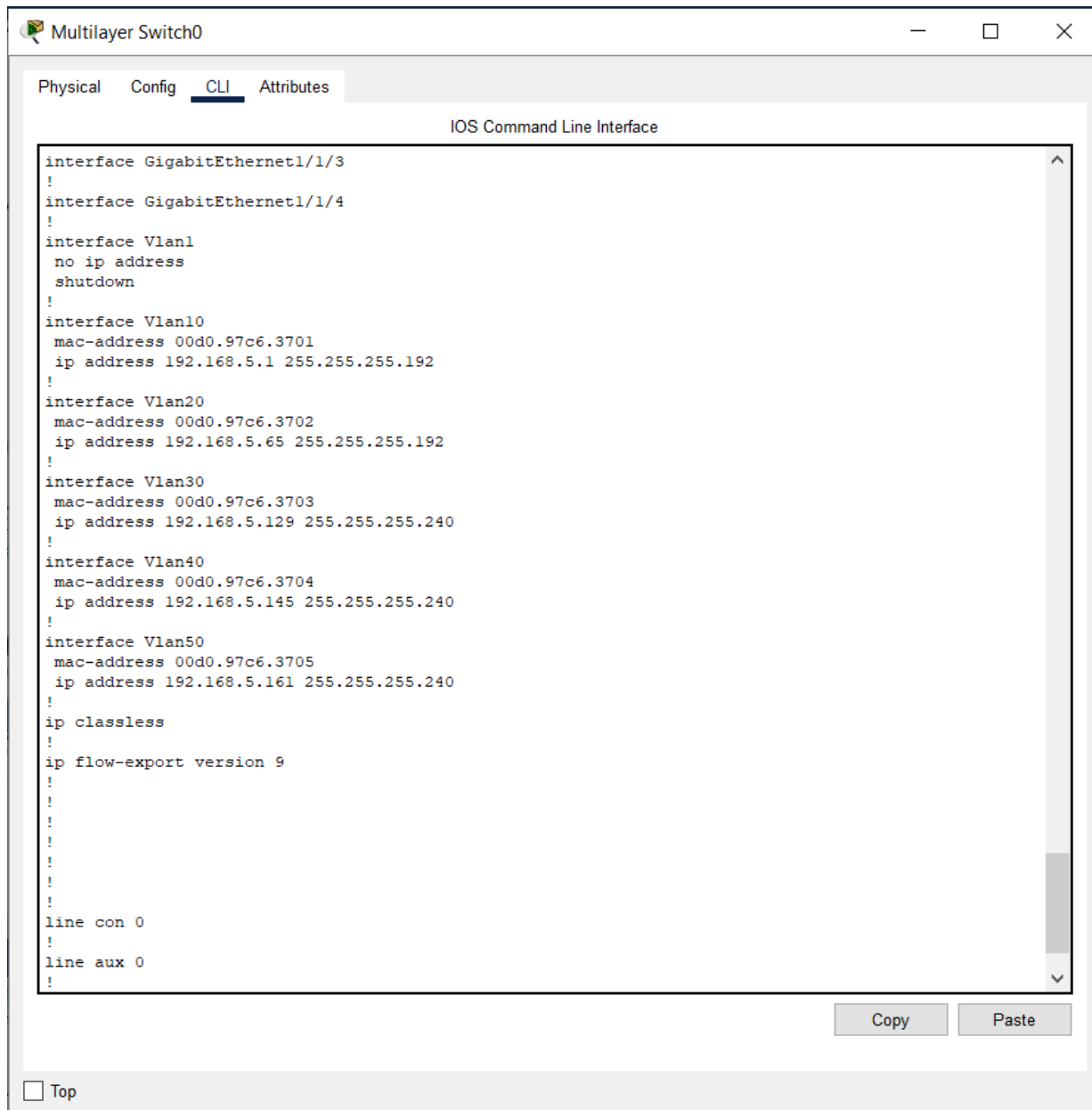
```
!
interface Port-channel1
!
interface GigabitEthernet1/0/1
  switchport trunk allowed vlan 10,20,30,40,50
  switchport mode trunk
!
interface GigabitEthernet1/0/2
  switchport trunk allowed vlan 10,20,30,40,50
  switchport mode trunk
!
interface GigabitEthernet1/0/3
  switchport trunk allowed vlan 10,20,30,40,50
  switchport mode trunk
!
interface GigabitEthernet1/0/4
!
interface GigabitEthernet1/0/5
!
interface GigabitEthernet1/0/6
  switchport trunk allowed vlan 10,20,30,40
  switchport mode trunk
  channel-group 1 mode active
!
interface GigabitEthernet1/0/7
  switchport trunk allowed vlan 10,20,30,40
  switchport mode trunk
  channel-group 1 mode active
!
interface GigabitEthernet1/0/8
!
interface GigabitEthernet1/0/9
!
interface GigabitEthernet1/0/10
!
interface GigabitEthernet1/0/11
!
interface GigabitEthernet1/0/12
!
interface GigabitEthernet1/0/13
!
interface GigabitEthernet1/0/14
!
```

Copy

Paste

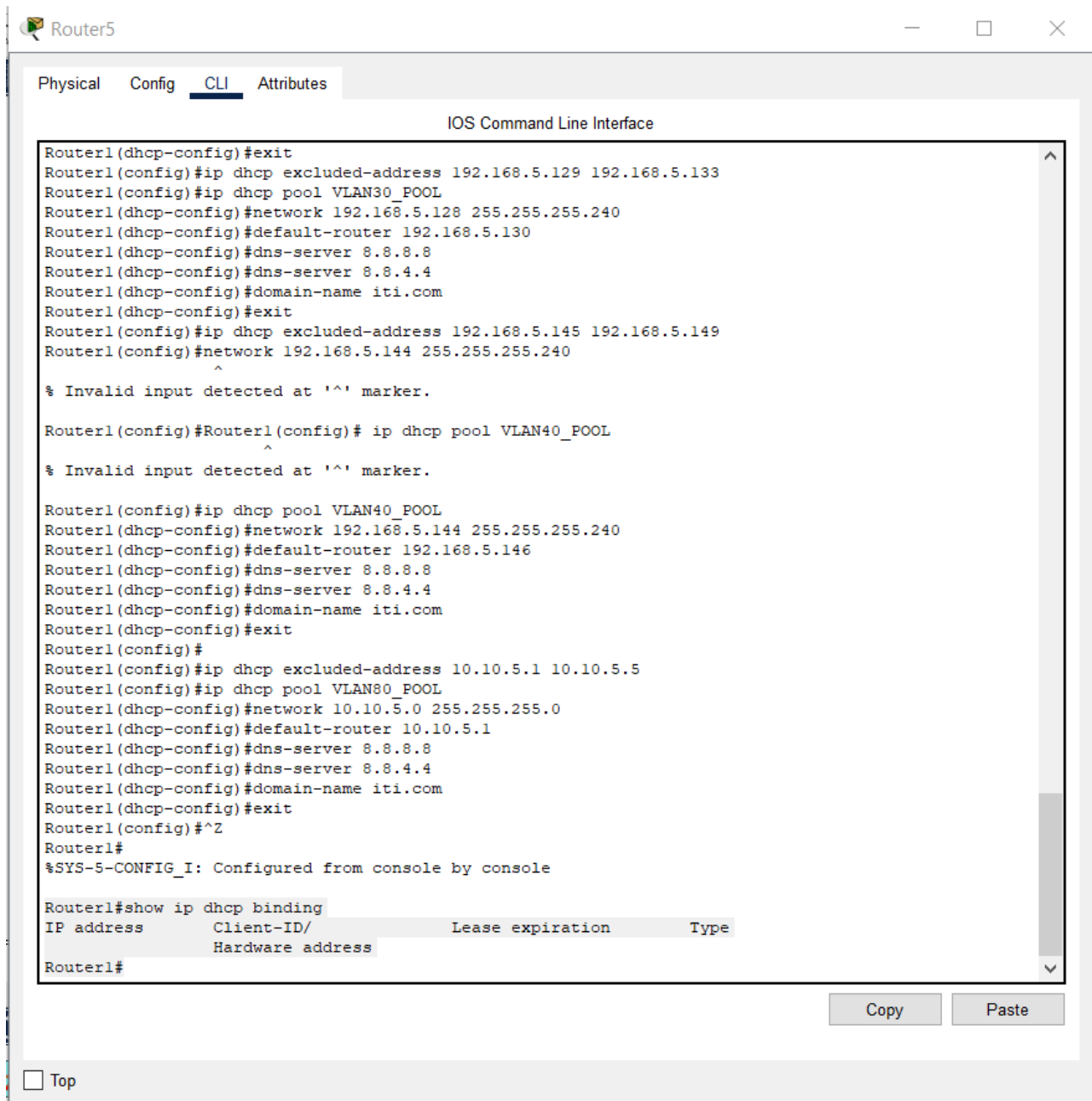
• VLANs

segment the Local Area Network (LAN) into smaller broadcast domains, isolating traffic for enhanced security and improved network performance



• DHCP

used to automatically distribute IP addresses, Default Gateway, and DNS server addresses to end devices (Hosts). Using a stand-alone server centralizes and simplifies IP address management



The screenshot shows a Cisco Router CLI interface with the following configuration steps:

```
Router1(dhcp-config)#exit
Router1(config)#ip dhcp excluded-address 192.168.5.129 192.168.5.133
Router1(config)#ip dhcp pool VLAN30_POOL
Router1(dhcp-config)#network 192.168.5.128 255.255.255.240
Router1(dhcp-config)#default-router 192.168.5.130
Router1(dhcp-config)#dns-server 8.8.8.8
Router1(dhcp-config)#dns-server 8.8.4.4
Router1(dhcp-config)#domain-name iti.com
Router1(dhcp-config)#exit
Router1(config)#ip dhcp excluded-address 192.168.5.145 192.168.5.149
Router1(config)#network 192.168.5.144 255.255.255.240
Router1(config)#ip dhcp pool VLAN40_POOL
Router1(dhcp-config)#network 192.168.5.144 255.255.255.240
Router1(dhcp-config)#default-router 192.168.5.146
Router1(dhcp-config)#dns-server 8.8.8.8
Router1(dhcp-config)#dns-server 8.8.4.4
Router1(dhcp-config)#domain-name iti.com
Router1(dhcp-config)#exit
Router1(config)#ip dhcp excluded-address 10.10.5.1 10.10.5.5
Router1(config)#ip dhcp pool VLAN80_POOL
Router1(dhcp-config)#network 10.10.5.0 255.255.255.0
Router1(dhcp-config)#default-router 10.10.5.1
Router1(dhcp-config)#dns-server 8.8.8.8
Router1(dhcp-config)#dns-server 8.8.4.4
Router1(dhcp-config)#domain-name iti.com
Router1(dhcp-config)#exit
Router1(config)#^Z
Router1#
%SYS-5-CONFIG_I: Configured from console by console

Router1#show ip dhcp binding
IP address      Client-ID/      Lease expiration    Type
              Hardware address
Router1#
```

At the bottom of the window, there are "Copy" and "Paste" buttons, and a "Top" button.

PC1

Physical

Config

Desktop

Programming

Attributes

IP Configuration

X

Interface

FastEthernet0

IP Configuration

☒ DHCP

☐ Static

DHCP request successful.

IPv4 Address

192.168.5.7

Subnet Mask

255.255.255.192

Default Gateway

192.168.5.2

DNS Server

8.8.4.4

IPv6 Configuration

☐ Automatic

☒ Static

IPv6 Address

/

Link Local Address

FE80::201:97FF:FE52:2734

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication

MD5

Username

Password

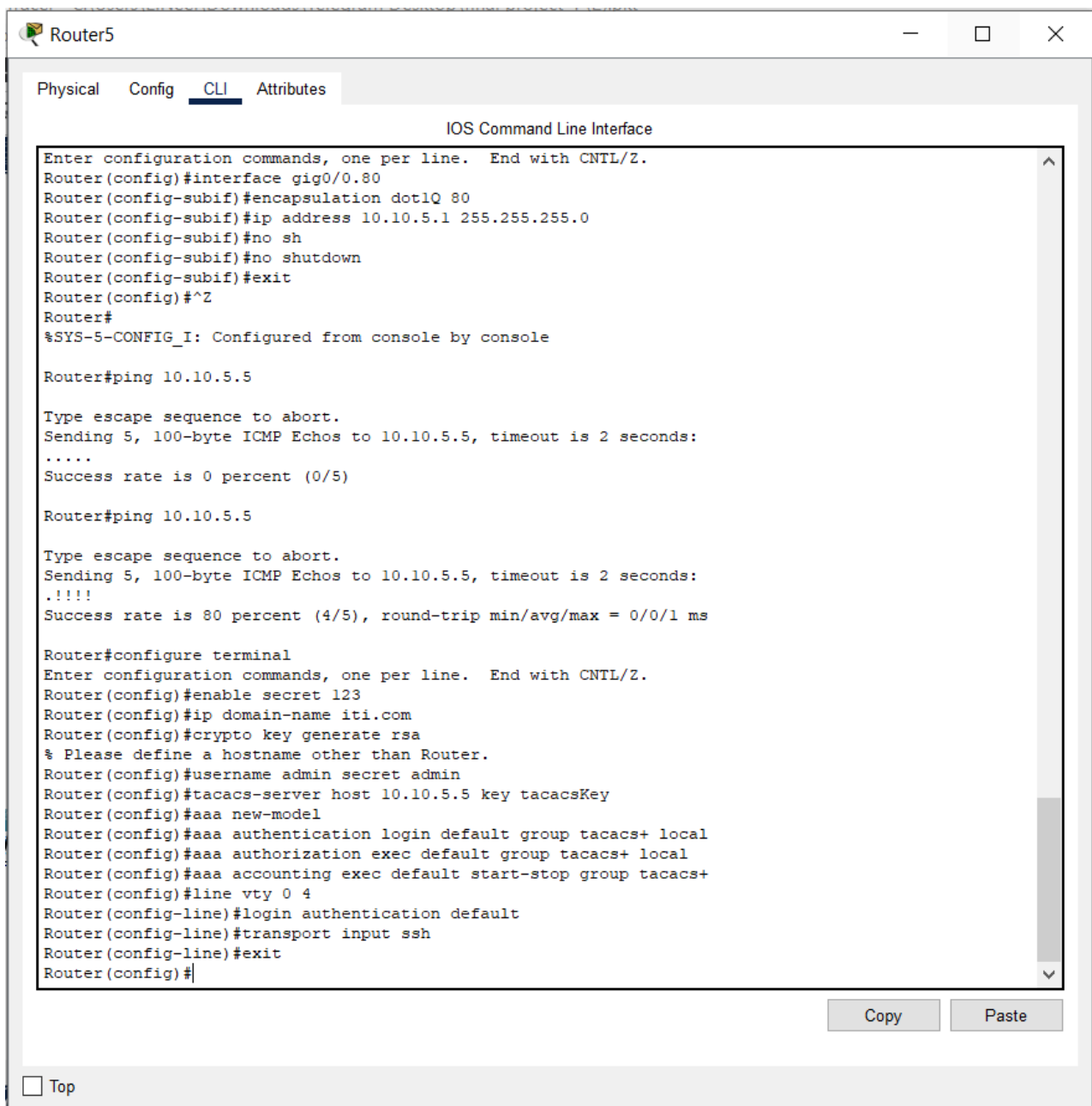
• AAA

Description: AAA (Authentication, Authorization, and Accounting) provides a secure and controlled method for accessing network devices

aaa new-model

aaa authentication login default group tacacs+ local

username admin secret admin



The screenshot shows a Cisco Router CLI interface with the following commands and output:

```
Router5
Physical Config CLI Attributes
IOS Command Line Interface

Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gig0/0.80
Router(config-subif)#encapsulation dot1Q 80
Router(config-subif)#ip address 10.10.5.1 255.255.255.0
Router(config-subif)#no sh
Router(config-subif)#no shutdown
Router(config-subif)#exit
Router(config)#^Z
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#ping 10.10.5.5

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.5.5, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)

Router#ping 10.10.5.5

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.5.5, timeout is 2 seconds:
..!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/1 ms

Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#enable secret 123
Router(config)#ip domain-name iti.com
Router(config)#crypto key generate rsa
% Please define a hostname other than Router.
Router(config)#username admin secret admin
Router(config)#tacacs-server host 10.10.5.5 key tacacsKey
Router(config)#aaa new-model
Router(config)#aaa authentication login default group tacacs+ local
Router(config)#aaa authorization exec default group tacacs+ local
Router(config)#aaa accounting exec default start-stop group tacacs+
Router(config)#line vty 0 4
Router(config-line)#login authentication default
Router(config-line)#transport input ssh
Router(config-line)#exit
Router(config)#
```

At the bottom of the window, there are "Copy" and "Paste" buttons, and a "Top" button with a checkbox.

Physical Config **Services** Desktop Programming Attributes**SERVICES**

HTTP

DHCP

DHCPv6

TFTP

DNS

SYSLOG

AAA

NTP

EMAIL

FTP

IoT

VM Management

Radius EAP

AAA

Service

☒ On ☐ Off

Radius Port

1645

Network Configuration

Client Name

Client IP

Secret

ServerType

Radius

	Client Name	Client IP	Server Type	Key
1	Router	10.10.5.1	Tacacs	123

Add

Save

Remove

User Setup

Username

Password

	Username	Password
1	network_admin	securepass

Add

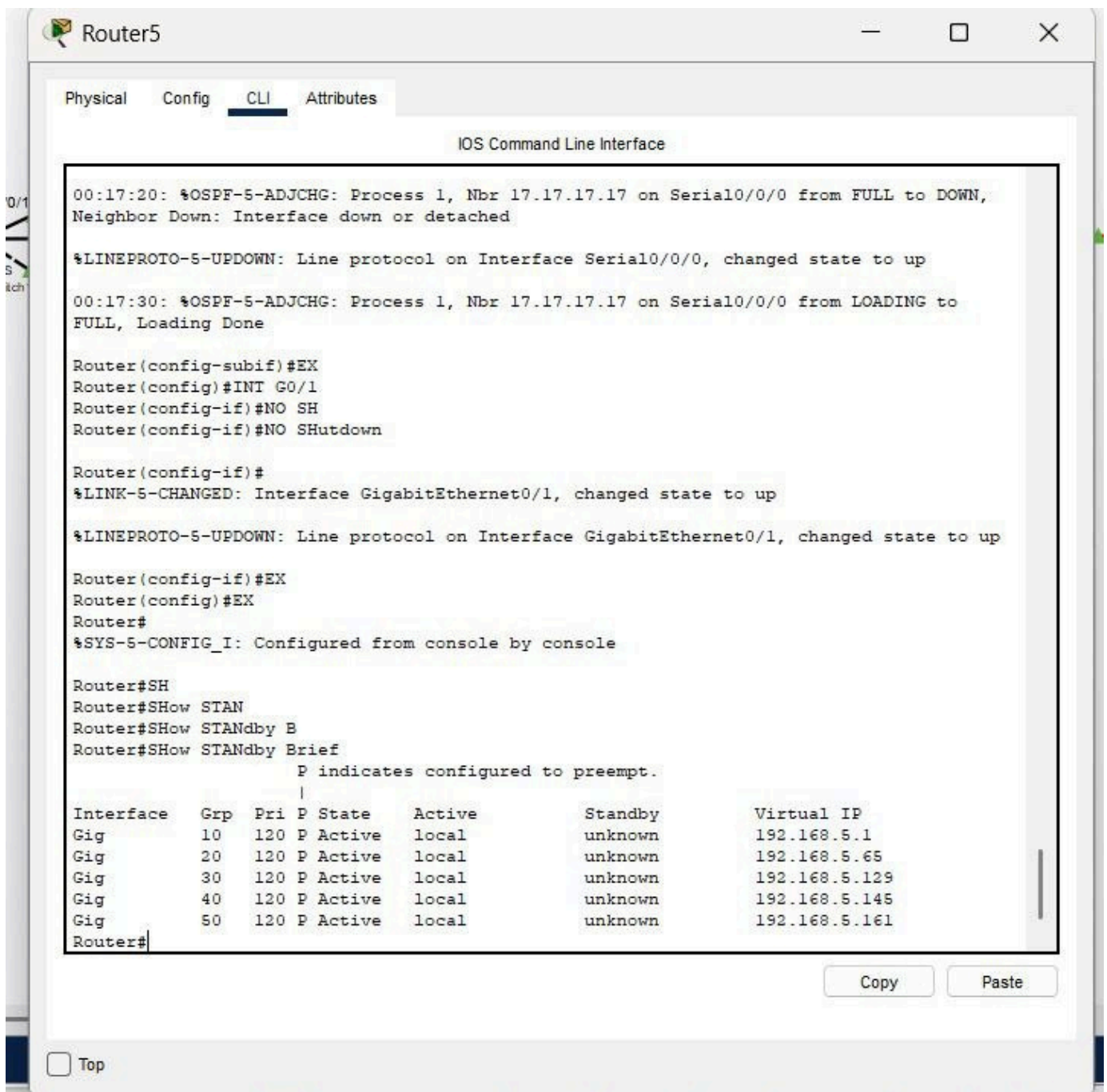
Save

Remove

• HSRP

It is used to ensure high network availability by providing automatic failover if the primary router fails.

- A single Virtual IP Address is shared between two routers (R1 and R2).
- One router is designated as Active, and the other is Standby.
- If the Active router fails, the Standby router instantly takes over the role of the default gateway without requiring host configuration changes



The screenshot shows the CLI of Router5 with the following output:

```
00:17:20: %OSPF-5-ADJCHG: Process 1, Nbr 17.17.17.17 on Serial0/0/0 from FULL to DOWN, Neighbor Down: Interface down or detached
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to up
00:17:30: %OSPF-5-ADJCHG: Process 1, Nbr 17.17.17.17 on Serial0/0/0 from LOADING to FULL, Loading Done

Router(config-subif)#EX
Router(config)#INT G0/1
Router(config-if)#NO SH
Router(config-if)#NO Shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Router(config-if)#EX
Router(config)#EX
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#SH
Router#SHoW STAN
Router#SHoW STANdby B
Router#SHoW STANdby Brief
      P indicates configured to preempt.
      |
Interface  Grp  Pri P State   Active      Standby      Virtual IP
Gig        10   120 P Active   local       unknown      192.168.5.1
Gig        20   120 P Active   local       unknown      192.168.5.65
Gig        30   120 P Active   local       unknown      192.168.5.129
Gig        40   120 P Active   local       unknown      192.168.5.145
Gig        50   120 P Active   local       unknown      192.168.5.161
Router#
```

Buttons for Copy and Paste are visible at the bottom right of the console window.

- **HSRP**

In an HSRP group, routers elect roles based on priority: one becomes Active, forwarding traffic, and the other becomes Standby, ready to take over instantly if needed. They share a common virtual MAC and virtual IP, so end devices always send traffic to the same logical gateway without knowing which physical router is handling it. HSRP uses periodic hello messages between routers to monitor availability. If the Standby router stops hearing the Active router's hellos, it automatically becomes the new Active, providing a smooth transition with minimal downtime. Traffic forwarding resumes quickly without requiring client reconfiguration or manual intervention. This redundancy model protects the first-hop connectivity for LAN and VLAN networks, eliminating single points of failure at the gateway level. It works alongside dynamic routing and link aggregation to strengthen both path redundancy and device reliability.

IOS Command Line Interface

```
Router(config-if)#
Router(config-if)#interface g0/0.10
Router(config-subif)# encapsulation dot1q 10
Router(config-subif)# ip address 192.168.5.2 255.255.255.192
Router(config-subif)# standby 10 ip 192.168.5.1
Router(config-subif)# standby 10 priority 120
Router(config-subif)# standby 10 preempt
Router(config-subif)#
Router(config-subif)#interface g0/0.20
Router(config-subif)# encapsulation dot1q 20
Router(config-subif)# ip address 192.168.5.66 255.255.255.192
Router(config-subif)# standby 20 ip 192.168.5.65
Router(config-subif)# standby 20 priority 120
Router(config-subif)# standby 20 preempt
Router(config-subif)#
Router(config-subif)#interface g0/0.30
Router(config-subif)# encapsulation dot1q 30
Router(config-subif)# ip address 192.168.5.130 255.255.255.240
Router(config-subif)# standby 30 ip 192.168.5.129
Router(config-subif)# standby 30 priority 120
Router(config-subif)# standby 30 preempt
Router(config-subif)#
Router(config-subif)#interface g0/0.40
Router(config-subif)# encapsulation dot1q 40
Router(config-subif)# ip address 192.168.5.146 255.255.255.240
Router(config-subif)# standby 40 ip 192.168.5.145
Router(config-subif)# standby 40 priority 120
Router(config-subif)# standby 40 preempt
Router(config-subif)#
Router(config-subif)#interface g0/0.50
Router(config-subif)# encapsulation dot1q 50
Router(config-subif)# ip address 192.168.5.162 255.255.255.240
Router(config-subif)# standby 50 ip 192.168.5.161
Router(config-subif)# standby 50 priority 120
Router(config-subif)# standby 50 preempt
Router(config-subif)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0.10, changed state to up
```

Copy

Paste

☐ Top

Scenario 0

Fire

Last Status

Source

Destination

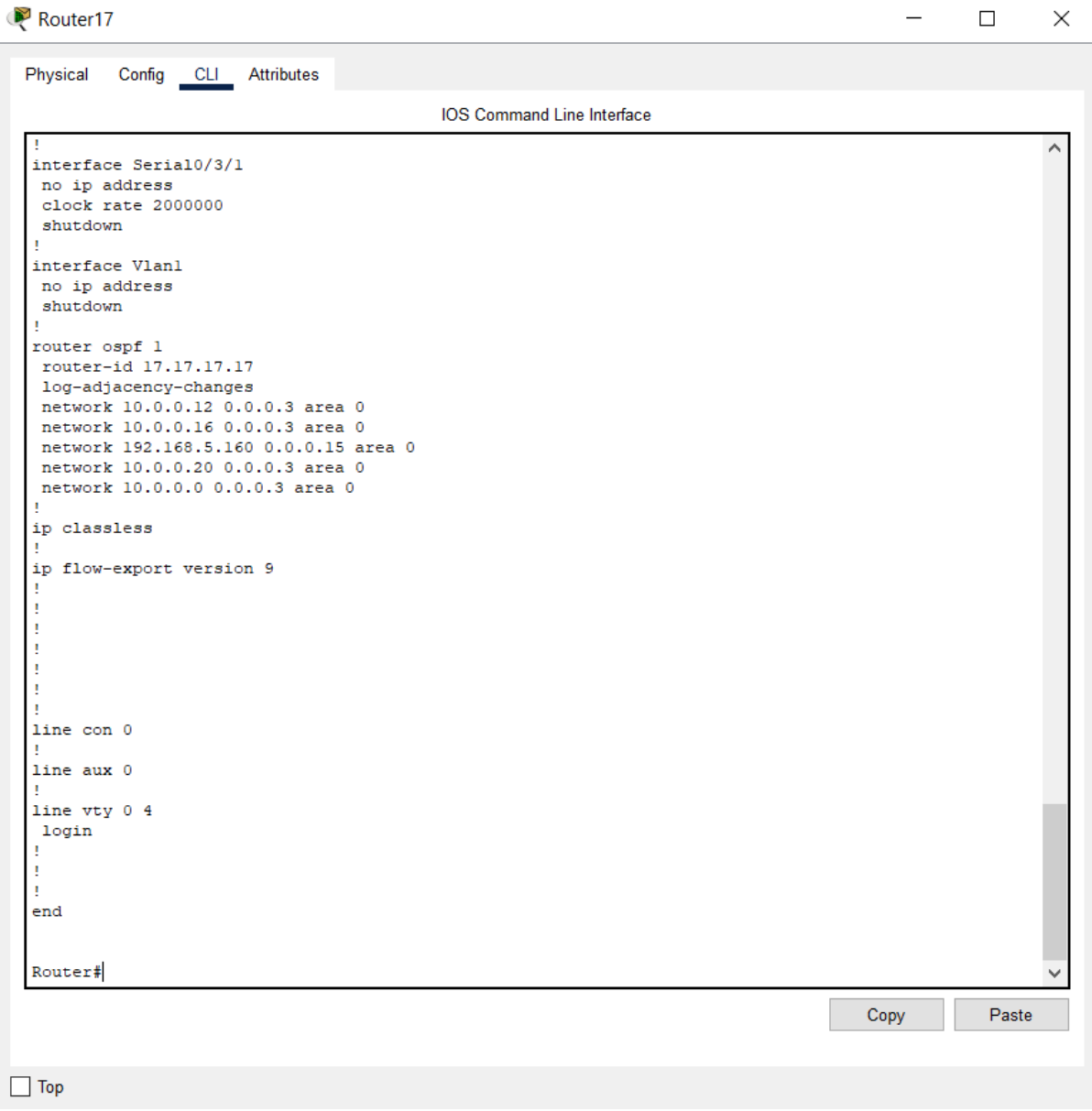
Type

Color

Time(sec)

• OSPF

I configured OSPF (Open Shortest Path First) as an Interior Gateway Protocol (IGP) that relies on the Link-State algorithm to determine the fastest paths based on calculated cost. We use OSPF to ensure that all routers within our internal network have comprehensive knowledge of all subnets (VLANs and Point-to-Point links) and can route traffic efficiently and quickly



```
!
interface Serial10/3/1
no ip address
clock rate 2000000
shutdown
!
interface Vlan1
no ip address
shutdown
!
router ospf 1
router-id 17.17.17.17
log-adjacency-changes
network 10.0.0.12 0.0.0.3 area 0
network 10.0.0.16 0.0.0.3 area 0
network 192.168.5.160 0.0.0.15 area 0
network 10.0.0.20 0.0.0.3 area 0
network 10.0.0.0 0.0.0.3 area 0
!
ip classless
!
ip flow-export version 9
!
!
!
!
!
!
!
line con 0
!
line aux 0
!
line vty 0 4
login
!
!
!
end

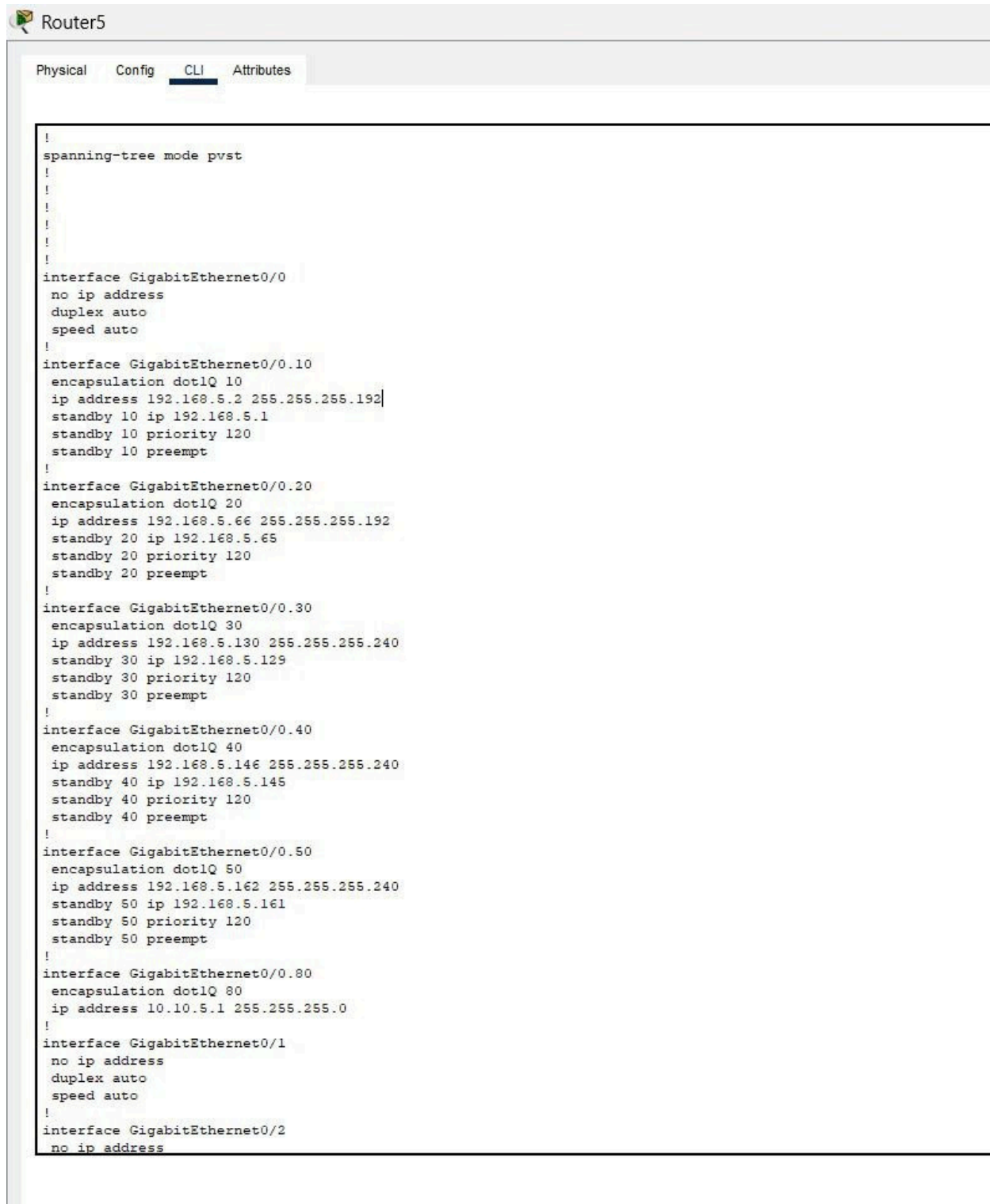
Router#
```

Copy Paste

☐ Top

• ROAS

I configured sub-interfaces on the router to enable inter-VLAN routing using the "Router on a Stick" method, allowing devices in different VLANs to communicate securely. It utilizes a single physical interface on the router (e.g., Gi0/0), which is logically divided into subinterfaces. Each subinterface is assigned to a specific VLAN using the **encapsulation dot1Q** protocol for tagging



```
!
spanning-tree mode pvst
!
!
!
!
!
!
!
interface GigabitEthernet0/0
no ip address
duplex auto
speed auto
!
interface GigabitEthernet0/0.10
encapsulation dot1Q 10
ip address 192.168.5.2 255.255.255.192
standby 10 ip 192.168.5.1
standby 10 priority 120
standby 10 preempt
!
interface GigabitEthernet0/0.20
encapsulation dot1Q 20
ip address 192.168.5.66 255.255.255.192
standby 20 ip 192.168.5.65
standby 20 priority 120
standby 20 preempt
!
interface GigabitEthernet0/0.30
encapsulation dot1Q 30
ip address 192.168.5.130 255.255.255.240
standby 30 ip 192.168.5.129
standby 30 priority 120
standby 30 preempt
!
interface GigabitEthernet0/0.40
encapsulation dot1Q 40
ip address 192.168.5.146 255.255.255.240
standby 40 ip 192.168.5.145
standby 40 priority 120
standby 40 preempt
!
interface GigabitEthernet0/0.50
encapsulation dot1Q 50
ip address 192.168.5.162 255.255.255.240
standby 50 ip 192.168.5.161
standby 50 priority 120
standby 50 preempt
!
interface GigabitEthernet0/0.80
encapsulation dot1Q 80
ip address 10.10.5.1 255.255.255.0
!
interface GigabitEthernet0/1
no ip address
duplex auto
speed auto
!
interface GigabitEthernet0/2
no ip address
```


• Nat

used to create a one-to-one, persistent mapping between an internal private IP address and a static external public IP address.

In this project, Static NAT is configured on Router R2 to allow external users (Internet) to consistently access our public Web Server. Any incoming request directed to the public IP address will be translated to the server's private IP address.

```
Router5
Physical Config CLI Attributes
IOS Command Line Interface
Enter configuration commands, one per line. End with CNTRL-Z.
Router(config)#interface g0/0
Router(config-if)# ip nat inside
Router(config-if)#
Router(config-if)#interface g0/1
Router(config-if)# ip nat outside
Router(config-if)#EX
Router(config)#ip nat inside source static 192.168.5.10 50.50.50.2
Router(config)#interface g0/1
Router(config-if)# ip address 50.50.50.1 255.255.255.252
Router(config-if)#
Router(config-if)#E
Router(config)#INT G0/2
Router(config-if)#NO SH
Router(config-if)#NO SHUTDOWN

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up

Router(config-if)#IP ADD
Router(config-if)#IP ADDRESS 50.50.50.2 255.255.0.0
% 50.50.0.0 overlaps with GigabitEthernet0/1
Router(config-if)#IP ADDRESS 50.51.50.2 255.255.0.0
Router(config-if)#ip nat outside
Router(config-if)#EX
Router(config)#interface g0/0
Router(config-if)# ip nat inside
Router(config-if)#EX
Router(config)#ip nat inside source static 192.168.5.10 50.50.50.2
Router(config)#ip nat inside source static 192.168.5.10 50.51.50.2
Router(config)#EX
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#
```

Copy Paste

☐ Top

Command Prompt

X

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 50.50.50.2

Pinging 50.50.50.2 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 50.50.50.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 50.51.50.2

Pinging 50.51.50.2 with 32 bytes of data:

Reply from 50.51.50.2: bytes=32 time<1ms TTL=255
Reply from 50.51.50.2: bytes=32 time=1ms TTL=255
Reply from 50.51.50.2: bytes=32 time<1ms TTL=255
Reply from 50.51.50.2: bytes=32 time<1ms TTL=255

Ping statistics for 50.51.50.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

- **SSH**

SSH provide secure device management.

```
ip domain-name iti.com
```

```
username admin secret password123
```

```
crypto key generate rsa
```

```
How many bits in the modulus [512]: 2048
```

```
line vty 0 4
```

```
transport input ssh
```

```
login local
```

```
exit
```

```
enable secret securepass
```


IOS Command Line Interface

```
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gig0/0.80
Router(config-subif)#encapsulation dot1Q 80
Router(config-subif)#ip address 10.10.5.1 255.255.255.0
Router(config-subif)#no sh
Router(config-subif)#no shutdown
Router(config-subif)#exit
Router(config)#^Z
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#ping 10.10.5.5

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.5.5, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)

Router#ping 10.10.5.5

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.5.5, timeout is 2 seconds:
..!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/1 ms

Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#enable secret 123
Router(config)#ip domain-name iti.com
Router(config)#crypto key generate rsa
% Please define a hostname other than Router.
Router(config)#username admin secret admin
Router(config)#tacacs-server host 10.10.5.5 key tacacsKey
Router(config)#aaa new-model
Router(config)#aaa authentication login default group tacacs+ local
Router(config)#aaa authorization exec default group tacacs+ local
Router(config)#aaa accounting exec default start-stop group tacacs+
Router(config)#line vty 0 4
Router(config-line)#login authentication default
Router(config-line)#transport input ssh
Router(config-line)#exit
Router(config)#
```

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